Spectral Deconvolution Algorithm for Global Fine-Mode Aerosol Retrieval in the 1990s Using Dual-Angle Satellite Aerosol Data

Xing Yan, Jiayi Chen, Chen Zuo, Yize Jiang, and Bin He

***Abstract*— Fine-mode aerosol optical depth (fAOD) is related to anthropogenic aerosols, which play an essential role in global climate change and the atmospheric environment. However, knowledge of the global fAOD in the 1990s is very limited. Compared with general satellite-based fAOD retrieval methods, the spectral deconvolution algorithm (SDA) provides an entirely different way to calculate fAOD. In this study, we derived three new SDA-based global daily fAOD (0.55μm) datasets from 1995 to 2003 with 1° × 1° spatial resolution using the second Along-Track Scanning Radiometer (ATSR-2) aerosol product data. In global-scale validation, the application of SDA to Swansea University (SU)–ATSR aerosol products for fAOD retrievals showed the best agreement with AErosol RObotic NETwork (AERONET) fAOD, with a correlation coefficient of 0.72, a root-mean-square error (RMSE) of 0.12, and ∼65.36% of retrievals inside of the estimated error envelope (EE) (±[0.05 + 15%]). Regionally, SDA improves retrieval performance in Southeast Asia and USA when using the SU and ATSR-dual view (ADV) aerosol products. However, the application of SDA to Oxford-RAL Aerosol and Cloud (ORAC)-based aerosol data considerably worsened its accuracy compared with the official ORAC fAOD. With the improved SU–SDA fAOD, the long-term trend showed a considerable decrease in fAOD in Central South America from 1995 to 2003. However, significantly increasing trends were observed in India and China. As SDA uses spectral AOD only, this study demonstrates that it is a promising method for optical satellite spatial fAOD retrieval.**

***Index Terms*— Fine-mode aerosol, global scale, satellite.**

I. Introduction

FINE-MODE aerosols have been an essential topic of study in recent decades because they are related to anthropogenic aerosols and have an important influence on global climate change, air quality, and human health [1], [2], [3]. Assessing the impact of fine-mode aerosols is challenging, especially when a very limited understanding of the global distribution of fine-mode aerosols is available [4], [5], [6]. Satellite observations offer an efficient solution to this challenge, wherein fine-mode aerosol amounts are represented as fine-mode aerosol optical depth (fAOD) [7]. Many satellite-based fAOD data have been widely used for regional and global studies, such as Moderate Resolution Imaging Spectroradiometer (MODIS) [8], [9], [10], Multi-angle Imaging SpectroRadiometer (MISR) [11], [12], [13], and POLarization and Directionality of the Earth's Reflectances (POLDER) [14], [15], [16]. However, these satellites only started providing data after 2000 and cannot offer fAOD information for the 1990s. Consequently, our global-scale understanding of fAOD before 2000 is restricted, which poses a significant obstacle to comprehending crucial, long-term aerosol trends essential for climate research. The 1990s, characterized by substantial industrial growth, particularly in countries such as China, held immense significance in environmental studies. Hence, obtaining fAOD data from that era is imperative for assessing historical changes, enhancing climate models, and informing sound decision-making.

The second Along-Track Scanning Radiometer (ATSR-2), a dual-view instrument, was launched onboard the European Remote Sensing in 1995 and provided data until 2003. It was one of the earliest satellites used for the retrieval of aerosol optical properties. Several ATSR-2-based aerosol products have been released and validated at global and regional scales. The European Space Agency Climate Initiative (CCI) Aerosol Project (Aerosol-cci) provides independently validated and high-quality algorithms for processing long-term records of aerosol properties from ATSR-2 (https://climate.esa.int/en/projects/aerosol/). Three main aerosol retrieval algorithms were used in Aerosol-cci for ATSR-2: Swansea University (SU)–ATSR [17], Oxford-RAL Aerosol and Cloud (ORAC) [18], and ATSR-dual view (ADV) [19]. These aerosol retrieval algorithms can retrieve both aerosol optical depth (AOD) and fAOD; however, currently, validation is always focused on AOD, and the assessment of fAOD is very limited [20], [21], [22].

In addition to satellites, ground-based measurements can offer fAOD information, such as the Aerosol Robotic Network (AERONET) [23]. AERONET provides fAOD using a spectral deconvolution algorithm (SDA) [24]. The SDA differs from satellite-based fAOD inversions that require aerosol mode

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assumptions, and it only requires AOD values for multiple channels (at a minimum of three wavelengths) [25]. Having SU, ORAC, and ADV retrieve AOD from multiple channels makes it possible to directly apply SDA to these three algorithms for ATSR-2 fAOD inversion. fAOD calculated from different algorithms may vary significantly, and there is no single method that performs best in all areas. Thus, accurate knowledge of spatially dependent differences is particularly crucial for global fAOD studies during the 1990s, especially over land, where most anthropogenic aerosols are located.

To fill this research gap, in this study, we addressed three fundamental questions: 1) does SDA have the potential to be used for satellite-based large-scale fAOD retrieval? 2) can SDA improve fAOD retrieval accuracy? and 3) how do SU, ORAC, ADV, and SDA differ for fAOD inversion. To solve these issues, we introduced SDA in ATSR-2 for global fAOD retrieval from 1995 to 2003. To further validate and compare different fAOD retrieval methods, the results were evaluated using ground-based measurements. Finally, spatial and temporal variations in fAOD and their tendencies over global cover during the study period were identified.

II. Methodology and Data

A. Aerosol-cci Products

We collected three Aerosol-cci Level 3 (L3) daily aerosol products retrieved by the SU–ATSR (version 4.3), ORAC (version 4.01), and ADV (version 2.31) algorithms from 1995 to 2003 (https://data.ceda.ac.uk/neodc/esacci/aerosol/data). The L3 aerosol data were aggregated over a $1^\circ \times 1^\circ$ grid from the L2 products (10×10 km^2), which included all the ATSR-2 orbits for each day. Besides AOD, all three aerosol retrieval algorithms provide individual fAOD at 0.55 μm, which were collected for evaluation and comparison purposes. Detailed descriptions of the principles and structure of the SU–ATSR, ORAC, and ADV algorithms can be found in their theoretical baseline documents (https://climate.esa.int/en/projects/aerosol).

B. AERONET fAOD

Ground-based observation data from AERONET are obtained as "truth" data for validating satellite-based retrieved fAOD. The fAOD (0.5 μm) and Ångström exponent of fAOD (AE–fAOD) were collected from the AERONET Version 3 Direct Sun Algorithm-based SDA Level 1.5 product (cloud-screened and quality controlled) [26]. Since AERONET does not directly provide the fAOD at 0.55 μm, we used the fAOD at 0.5 μm and AE–fAOD to calculate fAOD at 0.55 μm as follows [2]:

$$\tau_2 = \exp\left[\ln(\tau_1) - \ln\left(\frac{\lambda_2}{\lambda_1}\right) \times \alpha\right] \tag{1}$$

where τ_2 and τ_1 are fAOD at 0.55 and 0.5 μm, respectively; λ_2 and λ_1 are the 0.55- and 0.5-μm wavelengths, respectively, and α is AE-fAOD.

For the purpose of data validation, we collected the retrieved fAOD values that were closest to the AERONET coordinates [27]. We established a time window of ± 1 h and defined the estimated error envelope (EE) as $\pm[0.05 + 15\%]$ [28].

C. Application of SDA for ATSR-2 fAOD Retrieval

O'Neill et al. [25] and Giles et al. [29] developed an SDA for discriminating aerosol optical properties of fine- and coarse-mode components. This approach relies on AOD spectral derivatives to first derive the aerosol fine-mode fraction (FMF) and then obtain fAOD from fAOD = AOD × FMF. Giles et al. [29] first proposed the following simple expressions for FMF estimation:

$$\begin{cases} \alpha_f^1 = \dfrac{1}{2(1-a)}\left\{\left(\alpha - \alpha_c - \dfrac{\alpha' - \alpha_c'}{\alpha - \alpha_c} + b^*\right) + \left[\left(\alpha - \alpha_c - \dfrac{\alpha' - \alpha_c'}{\alpha - \alpha_c} + b^*\right)^2 + 4c^*(1-a)\right]^{1/2}\right\} + \alpha_c \\ \eta^1 = \dfrac{\alpha - \alpha_c}{\alpha_f^1} + \alpha_c \end{cases} \tag{2}$$

where α_f^1 and η^1 are the uncorrected estimates of the fine-mode AE and FMF, respectively; α is the AE; α_c is coarse-mode AE, which has a constant value of -0.15 in SDA; and α' and α_c' are the AE derivatives and coarse-mode AE derivatives, respectively ($\alpha_c' = 0$ in the SDA). α' is calculated as

$$\alpha' = -2\alpha_2 \tag{3}$$

where α_2 is derived using three AOD values (τ_1–τ_3) at specific wavelengths (λ_1–λ_3)

$$\begin{cases} \ln\tau_1 = \alpha_2(\ln\lambda_1)^2 + \alpha_1(\ln\lambda_1) + \alpha_0 \\ \ln\tau_2 = \alpha_2(\ln\lambda_2)^2 + \alpha_1(\ln\lambda_2) + \alpha_0 \\ \ln\tau_3 = \alpha_2(\ln\lambda_3)^2 + \alpha_1(\ln\lambda_3) + \alpha_0. \end{cases} \tag{4}$$

For ATSR-2 fAOD calculation, as shown in Fig. 1, we extract the AOD value at three wavelengths from Aerosol-cci products SU–ATSR (0.55, 0.67, and 0.87 μm), ORAC (0.55, 0.67, and 0.87 μm), and ADV-based (0.55, 0.67, and 1.6 μm) aerosol products. ADV has no AOD at 0.87 μm over land due to strong reflectance by vegetation at this wavelength. In addition, we also obtained the AE (α) from SU–ATSR (0.55–0.87 μm), ORAC (0.55–0.87 μm), and ADV-based (0.55–0.67 μm) aerosol products for the calculation of (2). In (2), the parameters b^* and c^* are

$$\begin{cases} b^* = b + 2\alpha_c a \\ c^* = c + (b + a\alpha_c)\alpha_c - \alpha_c'. \end{cases} \tag{5}$$

The expression of b^* was incorrect in [29] and the authors corrected this error in [30]. In addition, in contrast to [29], new spectrally dependent coefficients of the parameters a, b, and c were modified as follows [31]:

$$\begin{cases} a = (a_{\text{lower}} + a_{\text{upper}})/2 \\ b = (b_{\text{lower}} + b_{\text{upper}})/2 \\ c = (c_{\text{lower}} + c_{\text{upper}})/2 \end{cases} \tag{6}$$

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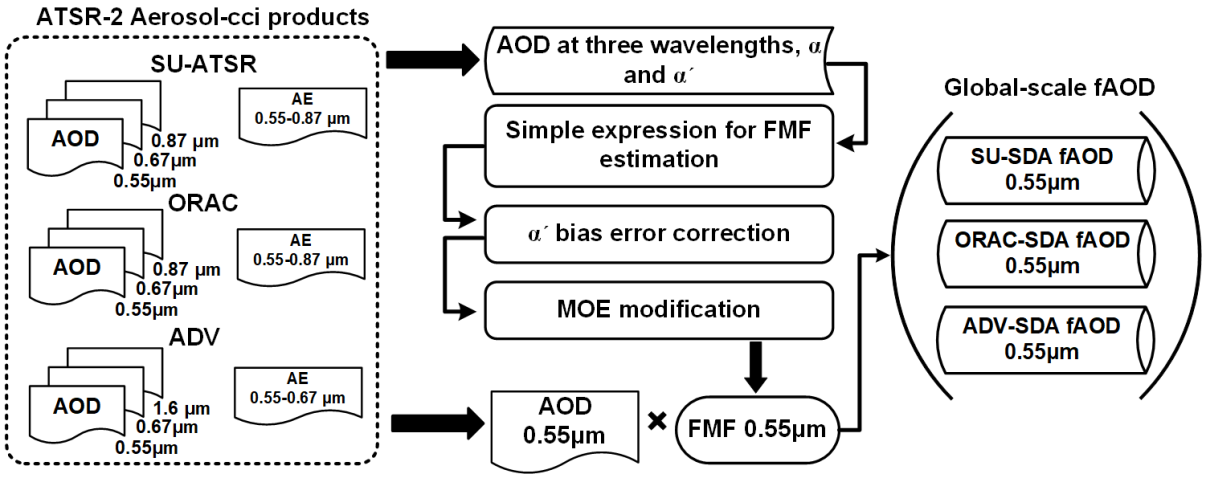


Fig. 1. Technical flowchart for the production of three SDA-based global fAODs by ATSR-2 Aerosol-cci products.

where

$$\begin{pmatrix} a_{\text{upper}} = -0.22, & a_{\text{lower}} = -0.3 \\ b_{\text{upper}} = 10^{-0.2388}\lambda^{1.0275}, & b_{\text{lower}} = 0.8 \\ c_{\text{upper}} = 10^{0.2633}\lambda^{-0.4683}, & c_{\text{lower}} = 0.63 \end{pmatrix}.$$

Incorporating (3), (5), and (6) into (2), all the parameters are known; thus, α_f^1 and η^1 can be calculated. However, the initially derived α_f^1 and η^1 contain errors based on (2), and the α' bias error correction was applied to overcome this issue [25]

$$\alpha'_{\text{error}} = 0.65 \times \exp[-(\eta^1 - 0.78)^2/(2 \times 0.18^2)]. \quad (7)$$

Then, the corrected α' ($\alpha'_{\text{corrected}}$) is

$$\alpha'_{\text{corrected}} = \alpha' + \alpha'_{\text{error}}. \quad (8)$$

By applying (8) to (2), we obtained the corrected $\alpha_f(\alpha_{f_{\text{corrected}}})$ and $\eta(\eta_{\text{corrected}})$. Then, a mean of extreme (MOE) modification (a smooth transition in the forcing condition $\eta \leq 1$) was employed in $\eta_{\text{corrected}}$ to calculate the final η [31].

III. Results

A. fAOD Validation and Comparison

Fig. 2 presents fAOD scatterplots, with each of the six datasets individually validated against AERONET fAOD at the 0.55-μm wavelength from 1995 to 2003. The smallest number of co-locations (3770 matched data points) was found in the ADV–SDA and official ADV fAOD. This is because ADV fails to provide adequate aerosol retrieval over bright surfaces [32]. Among the six fAOD results, SU–SDA showed the best performance, yielding an R of 0.72 and a root-mean-square error (RMSE) of 0.12. By incorporating SDA, there were improvements in the ADV–SDA and SU–SDA products, with more retrievals falling within the EE envelope than the ADV and SU fAOD. When applying the SDA to the ORAC product, its fAOD accuracy became considerably worse, whereas the official ORAC fAOD showed a better correlation ($R = 0.62$) with more retrievals within the EE (60.75%) and fewer errors (RMSE = 0.14). The reason for the decline in ORAC–SDA fAOD could be the error of α and α' in the ORAC AODs calculation in (2). The reliabilities of α and α' are sensitive to AOD accuracy at selected wavelengths (the errors in α and α' arise from the uncertainties in ORAC AOD at three wavelengths: 0.55, 0.67, and 0.87 μm). A previous study indicated that the uncertainty in the AOD at each wavelength can have a clear effect on FMF estimation, contributing to the uncertainty of the final fAOD calculation [33], [34]. Surprisingly, official ADV, SU, and ORAC fAOD products presented an overestimation issue (retrievals above the EE envelope by ~24%–28%); however, overestimation clearly decreased in ADV–SDA fAOD, except for SU–SDA and ORAC–SDA fAOD.

Fig. 3 shows the overall global correlation of fAOD at 0.55 μm for six datasets compared with AERONET fAOD from 1995 to 2003. According to the first two columns, in Southeast Asia (black circle), the correlation (R) between the retrieved fAOD and AERONET fAOD was low. However, the third column illustrates positive differences in the circled regions, which implies an improvement in correlation by applying SDA to the ATSR-2 Aerosol-cci products. Furthermore, despite the high station-based correlation in USA, an improvement in correlation (the third column, red square) was discovered in the SU–SDA and ADV–SDA products. In contrast, the deterioration of R emerged over South America (blue circle in the third column) after applying the SDA to ORAC products. Noticeably, in South Africa (black square), R was high in SU and ADV products (nearly >0.8), but low in ORAC products (nearly <0.3). In the first and second columns of Fig. S1 (as shown in the Supplementary Material), all six datasets demonstrate elevated RMSE values within the Amazon Plain region (indicated by the black box), contrasting with lower RMSE values found within USA (marked by the black circle). Moving to the third column, both SU–SDA and ADV–SDA notably diminish RMSE values across both USA and Europe. However, the ORAC-SDA product displays greater RMSE values within this region in contrast to the official data, suggesting that the implementation of SDA into the ORAC product does not yield a substantial improvement in RMSE reduction.

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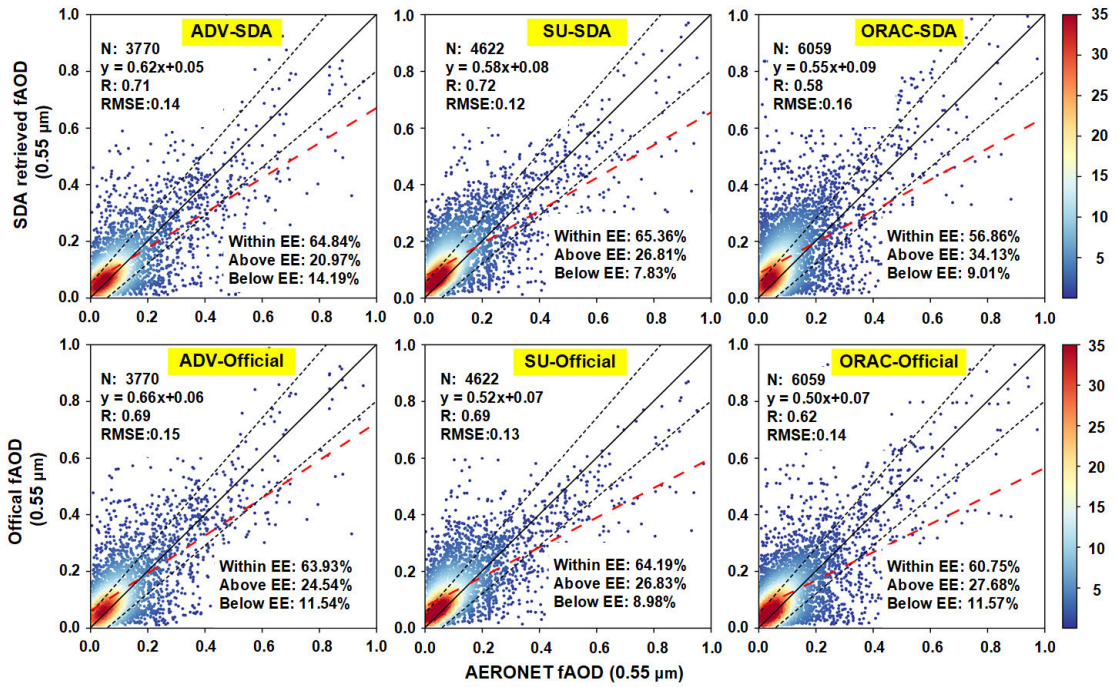


Fig. 2. Retrieved fAOD (at 0.55 μm) compared with 2-h averaged AERONET fAOD (average from 1 h before to 1 h after ATSR-2 satellite overpass time).

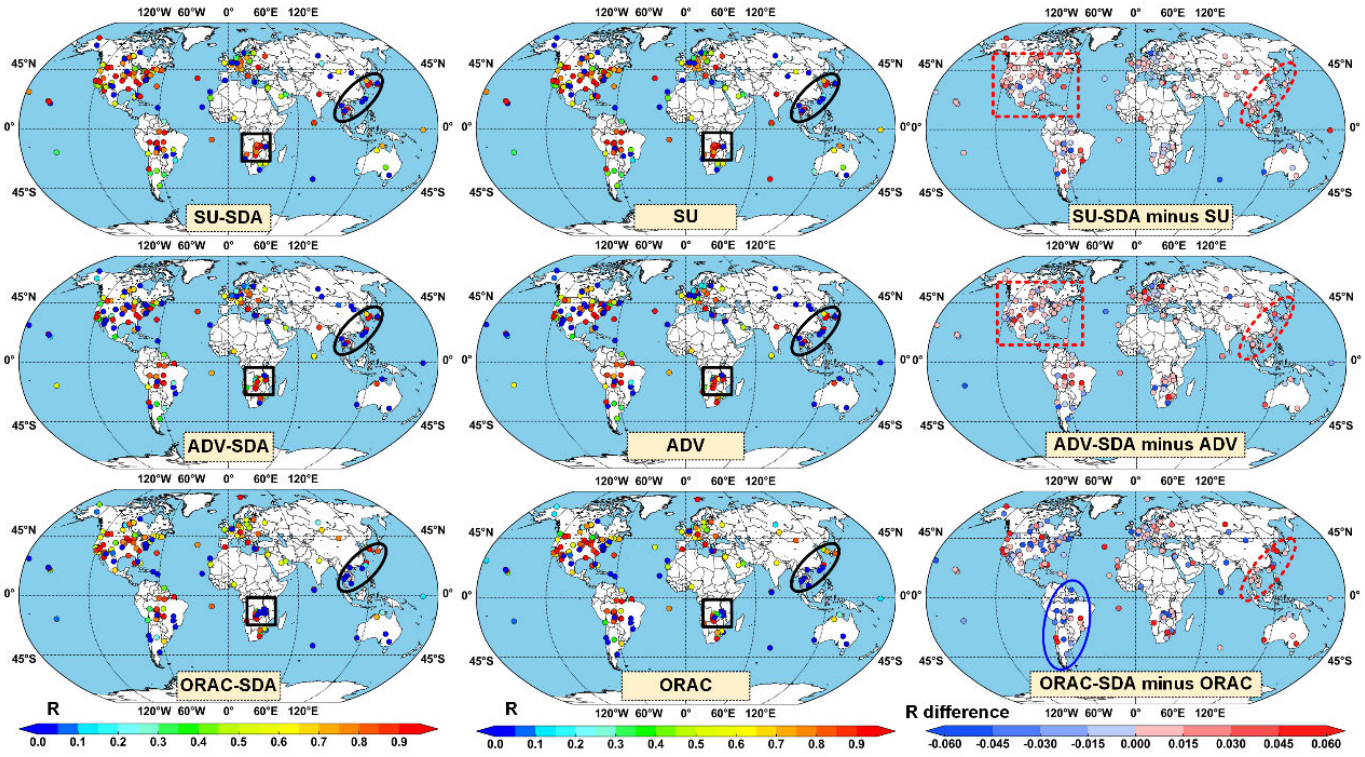


Fig. 3. Correlation of fAOD (at 0.55 μm) over land for six datasets compared with AERONET fAOD from 1995 to 2003. For each row, the 1st panel is retrieval ATSR-2 fAOD using SDA, the 2nd is the official ATSR-2 fAOD, and the 3rd panel is the differences between the SDA and official fAOD.

The annual fAOD derived by the SU–SDA, ADV–SDA, and ORAC–SDA from 1995 to 2003 is shown in the Supplementary Material, Figs. S2–S4. Generally, northern South America, eastern China, and southern Asia were detected as hotspots of fAOD in all SDA-based results. Their spatial differences from the official data are presented in Fig. 4(a)–(c). From the comparison, the SU–SDA and SU products differed widely over most global lands, where the values of SU–SDA were higher than those of SU, especially in eastern China, northeastern India, central Africa, and the Amazon. In contrast, fAOD values were lower for ADV–SDA than for ADV, especially along some coastal regions, such as the coastal areas

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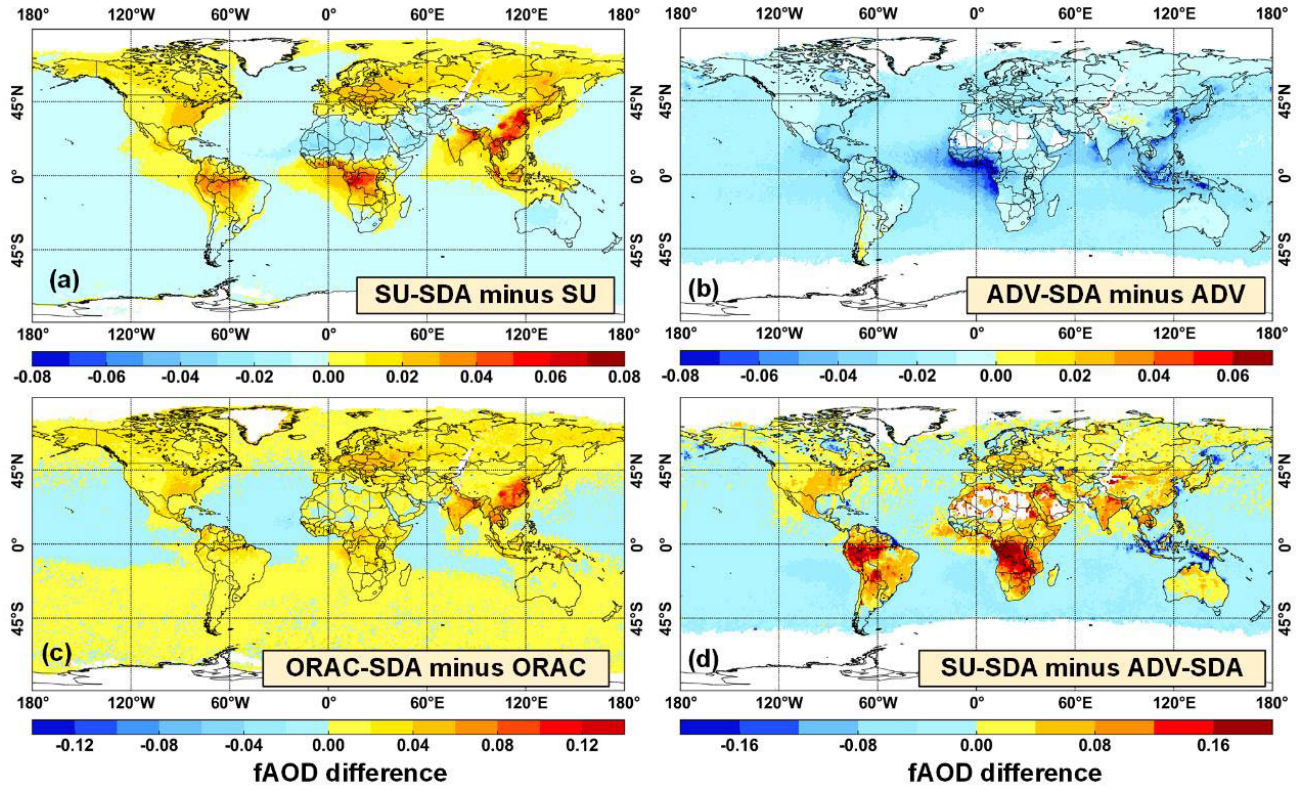


Fig. 4. Gridded differences between averaged 1° × 1° fAOD (at 0.55 μm) retrieved from (a) SU–SDA and SU, (b) ADV–SDA and ADV, (c) ORAC–SDA and ORAC, and (d) SU–SDA and ADV–SDA during 1995–2003.

of western Africa, eastern China, and northern South America. The differences between ORAC–SDA and ORAC were similar to those of the SU products; however, ORAC–SDA values were higher than those of ORAC extending to the ocean. Comparing the two best products (SU–SDA and ADV–SDA), we found that the difference between them on land was more evident than that on the ocean. The fAOD values of SU–SDA were dramatically higher than those of ADV–SDA, especially over southern Africa and northern South America. From Fig. 4, compared to the SU fAOD, the SU–SDA approach has mitigated the underestimation of fAOD near coastal areas. This improvement is also evident in the ORAC–SDA fAOD results. However, in the application of ADV data, the underestimation of fAOD by ADV–SDA near coastal areas is even more pronounced. This is primarily attributed to the high surface reflectance of coastal regions (due to factors such as pollution), which results in an unstable AOD calculation by ADV over highly reflective surfaces.

B. Global Distribution of fAOD and Trends From 1995 to 2003

From the validation, SU–SDA exhibited the best performance for global fAOD retrieval. Fig. 5(a) and (b) shows the spatial distribution of the worldwide mean SU–SDA fAOD for 1995–1999 and 2000–2003, respectively. High fAOD values (>0.4) were observed in eastern China and northern India during 1995 and 1999, which subsequently expanded over the following four years [see Fig. 5(a) and (b)]. This is evident in Fig. 5(c), with statistically significant positive trends of >0.015/year in northern India and 0.02/year ($p < 0.1$) in eastern China from 1995 to 2003. This is claimed to correspond to rapid urbanization and industrial development [35], [36], [37]. In contrast, although fAOD values over central South America were also high [>0.35, Fig. 5(a)] during 1995–1999, they were lower during 2000–2003 [see Fig. 5(b)], with a statistically significant negative trend detected from 1995 to 2003 [<−0.015/year in Fig. 5(c), p-value < 0.1]. This discovery extends the findings of Bevan et al. [38], who observed a declining trend in AOD from 1995 to 2000, which was associated with wildfire occurrences. Notably, during the period spanning from 1995 to 2003, western USA demonstrated statistically significant positive trends in fAOD, as shown in Fig. 5(c). In a related investigation, a quantile regression analysis uncovered an upward trajectory in $PM_{2.5}$ (a major component of fAOD) within western USA between 1988 and 2016. This increase is likely attributable to wildfires, a trend that aligns seamlessly with our own results [39]. Interestingly, this phenomenon aligns with the observations of Hammer et al. [40], who likewise documented a similar pattern between 1998 and 2018. Moreover, high fAOD (>0.6) also occurred in Central Africa during both 1995–1999 and 2000–2003, but the trend of the corresponding regions was complex, with some areas showing an increasing trend and some showing a decreasing trend [see Fig. 5(c)]. As shown in Fig. 5(d), we compared fAOD trends between SU–SDA and AERONET at eight stations and discovered that almost all were consistent in trend and magnitude, especially in New

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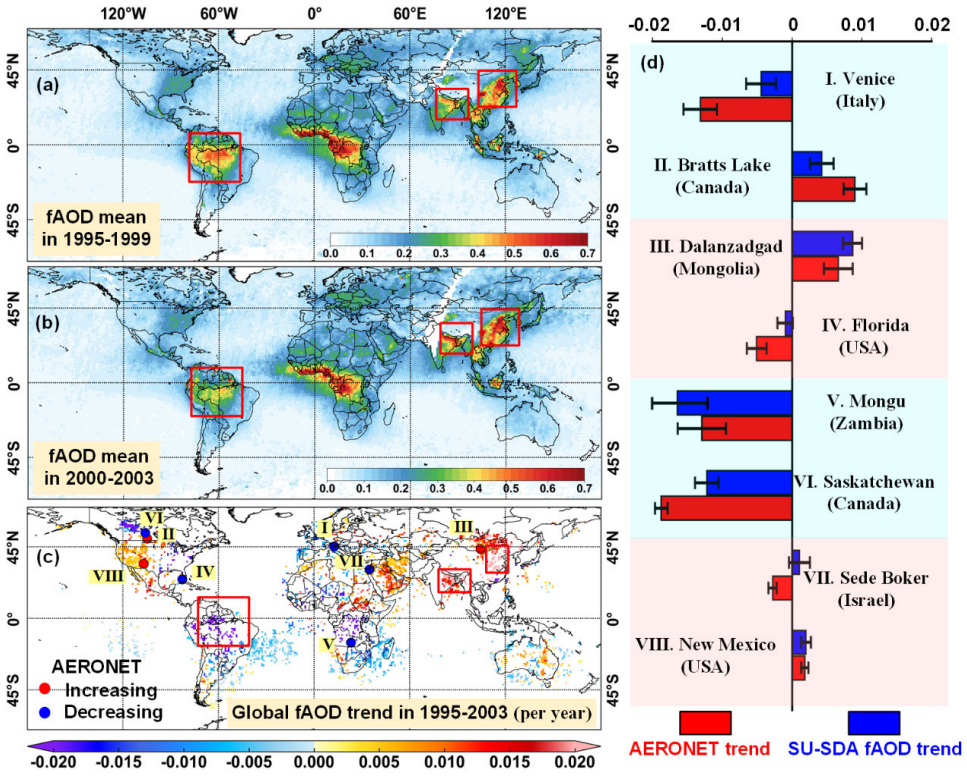


Fig. 5. Spatial distribution of averaged fAOD (at 0.55 μm) for SU–SDA during (a) 1995–1999 and (b) 2000–2003. (c) Global annual trends during 1995–2003 and (d) corresponding annual trends of eight stations from AERONET and SU–SDA with statistical significance (p-value < 0.1). Black error bars represent five times standard deviation.

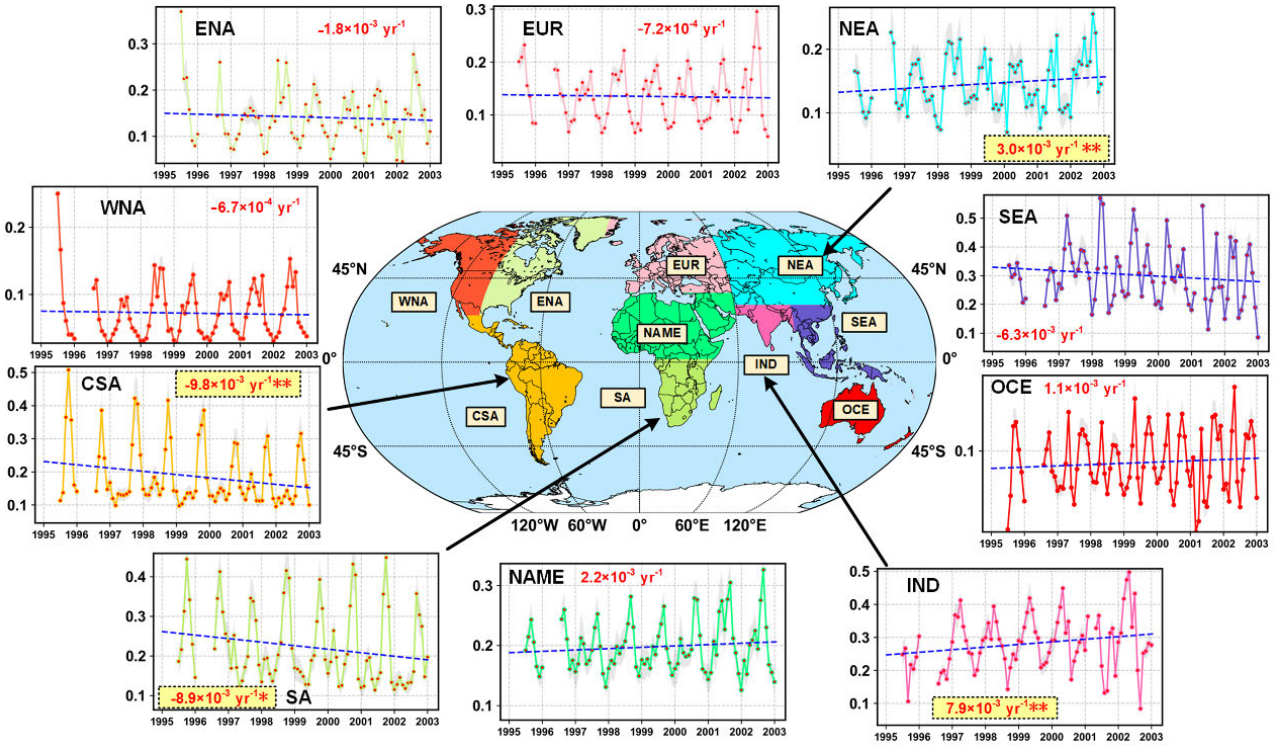


Fig. 6. Time series of fAOD over ten regions (NEA: North-East Asia, SEA: South-East Asia, OCE: Oceania, IND: India, NAME: North Africa/Middle East, SA: Southern Africa, CSA: Central/South America, WNA: Western North America, ENA: Eastern North America, and EUR: Eurasia) during 1995–2003, where colors of regions correspond to colors of lines. Shaded parts represent the monthly mean value plus/minus 10% standard deviation. Dashed lines indicate the fitting lines. "*" implies a statistically significant trend at a 90% significance level, and "**" implies a statistically significant trend at a 95% significance level.

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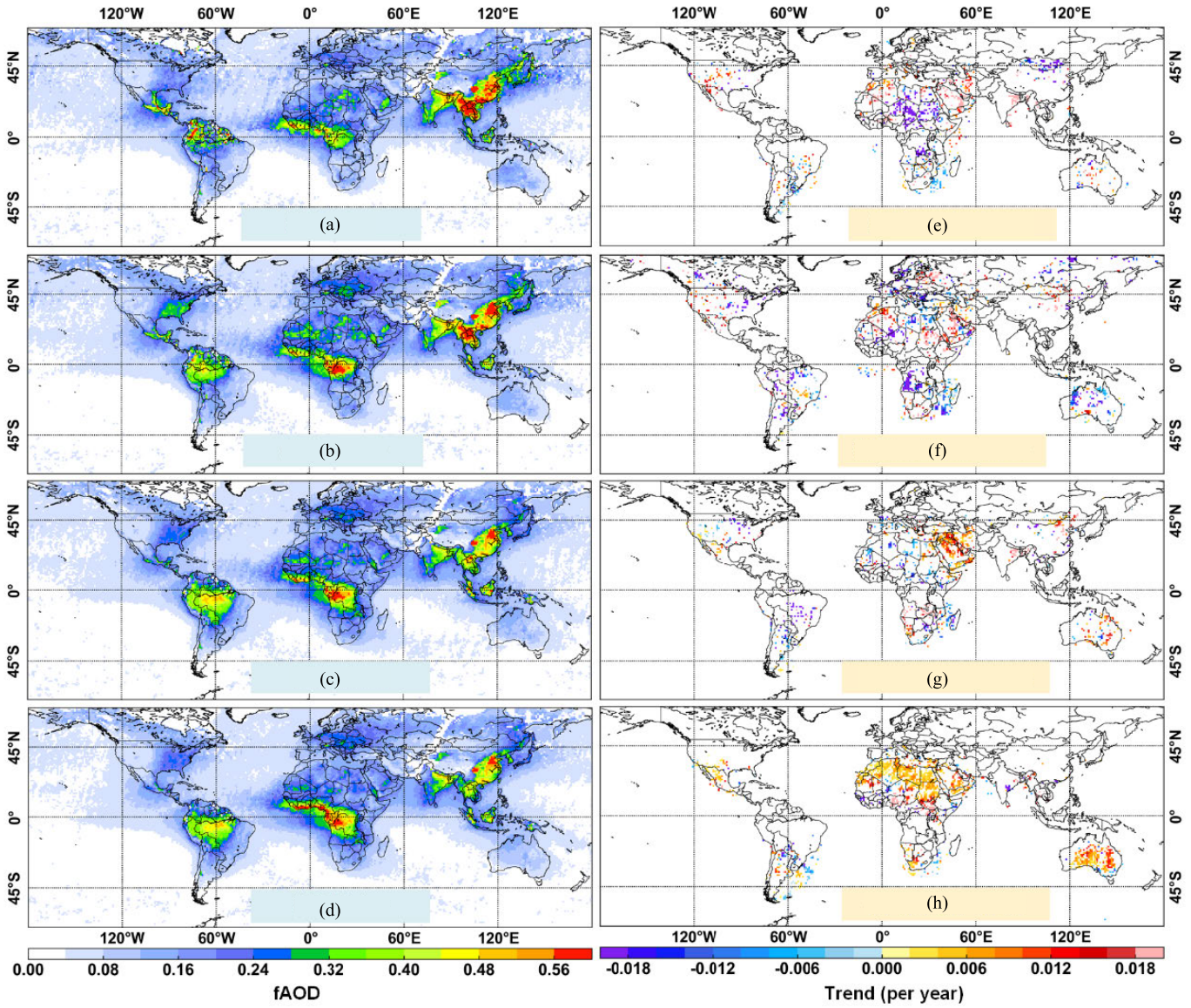


Fig. 7. Spatial distribution of global fAOD (at 0.55 μm) in (a) spring (March, April, and May), (b) summer (June, July, and August), (c) autumn (September, October, and November), and (d) winter (December, January, and February); and (e)–(h) fAOD seasonal trends in the corresponding seasons during 1995–2003 with statistically significant (p-value < 0.1).

Mexico in the USA. However, at Sede Boker (Israel), fAOD showed opposite trends in SU–SDA and AERONET.

Fig. 6 shows the monthly time series and fitting linear trends of fAOD in ten global regions from 1995 to 2003. The fAOD displayed declining trends in six regions and increasing trends in the other four regions, which were typically concentrated in the Eastern Hemisphere. The fAOD in India and Northeast Asia showed statistically significant positive trends of 7.9×10^{-3}/year and 3.0×10^{-3}/year, respectively, (95% significance level). Nevertheless, statistically significant negative trends were found in central South America ($p < 0.05$) and southern Africa ($p < 0.1$) of -9.8×10^{-3}/year and -8.9×10^{-3}/year, respectively. Although fAOD was found to be declining substantially in Europe, eastern North Asia, and western North Asia, and increasing in North Africa and Oceania, these trends did not pass the 90% significance test.

Fig. 7 shows the spatial distribution of seasonal variation in global fAOD at 0.55 μm (a)–(d) and their corresponding annual trends (e)–(h). Yunfeng et al. [41] found that AOD values were higher in spring over China during 1961–1990; here, we found that fAOD, which is part of AOD, was high in both spring and summer in China [see Fig. 7(a) and (b)] in 1995–2003. Moreover, fAOD values were higher in summer than in the other seasons over eastern USA [see Fig. 7(a)–(d)], which is consistent with findings that AOD and $PM_{2.5}$ concentrations have evident seasonality with higher values in the summer [41], [42]. During the summer months, a notable increase in fAOD levels was observed across eastern USA. This phenomenon can be attributed to the local seasonal production of sulfate aerosols as well as sporadic emissions from fires [42]. South America and Africa showed similar seasonal patterns, with higher fAOD values in summer, autumn, and winter, but lower values in spring. The spatial distribution

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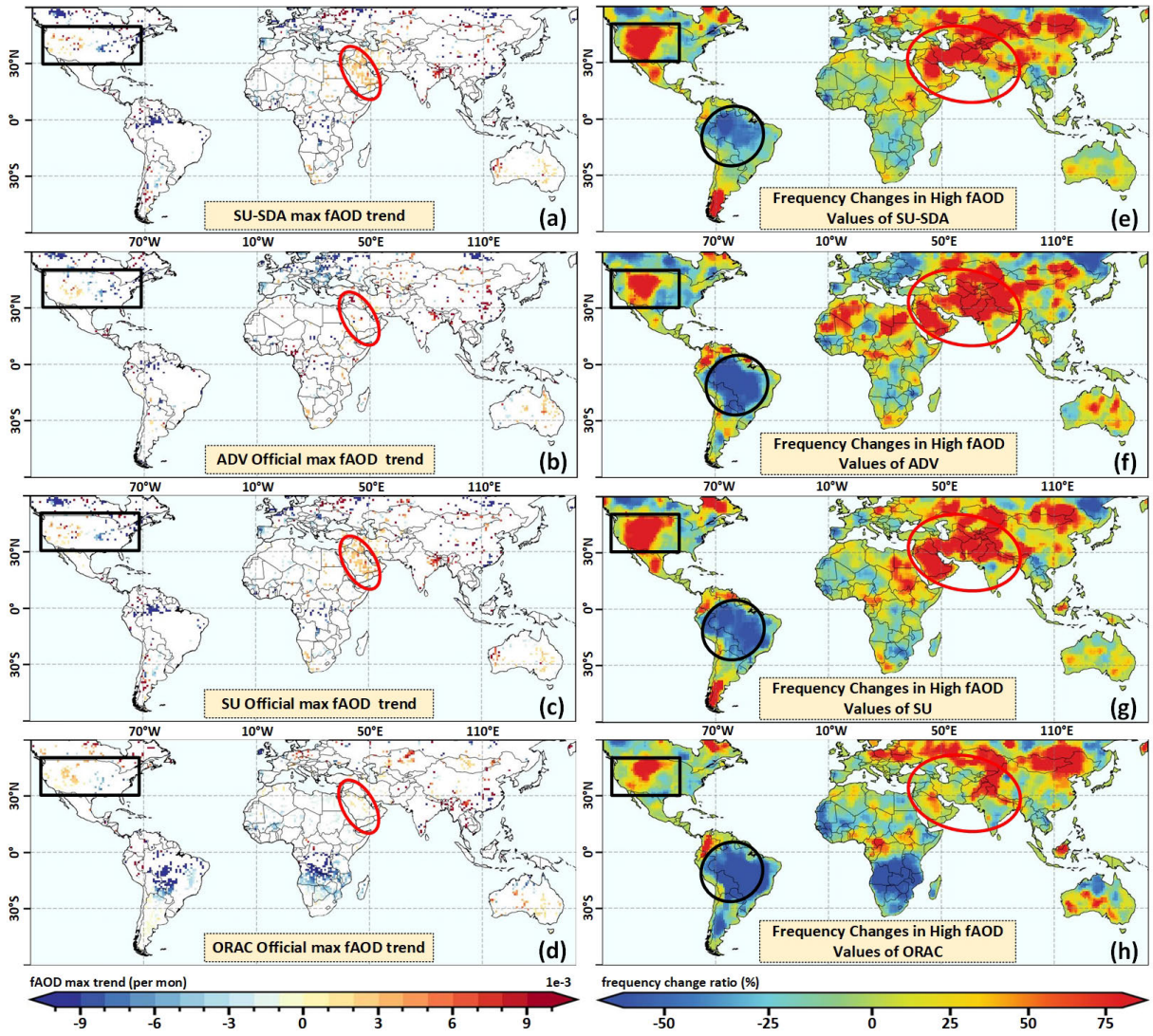


Fig. 8. Long-term patterns and shifts in the occurrence rate of unusual events involving the higher fAOD (at 0.55 μm, >1 standard deviation above the average) for four datasets (SU-SDA, SU, ADV, and ORAC) from 1995 to 2003. (a)–(d) Spatial distribution of the statistically significant trend of monthly maximum fAOD (at 0.55 μm) values in four datasets during 1995–2003 (p-value < 0.1). (e)–(h) Change in the number of days with high fAOD value divided by the total number of days between 2000 and 2003, in contrast to the anomalies recorded from 1995 to 1999.

of the annual seasonal trends is shown in Fig. 7(e)–(h) for 1995–2003 (p-value < 0.1). Statistically significant positive trends were exhibited over Australia in the winter (DJF). These trends could potentially be attributed to the occurrence of severe drought conditions in 2002, coupled with wildfires in January 2003, and an overall rise in biomass burning [43], [44]. Mitchell et al. [43] also found that this upward trend is more likely to occur during the season of intensified dust activity, especially from September to March, which is consistent with our findings. A similar seasonality was shown in Mexico, where statistically significant positive trends appeared in winter, which could correspond to increasing human activities, such as residential heating [45].

Fig. 8(a)–(d) shows statistically significant trends ($p < 0.1$) in the maximum monthly fAOD values across four datasets from 1995 to 2003. Evidently, all datasets display prominent upward trends in western USA while revealing downward trends in the eastern region (indicated by black boxes). In addition, statistically significant increasing trends in the minimum monthly fAOD values across USA are evident [see Fig. S5(a)–(d), see the Supplementary Material]. Moreover, compared to ADV and ORAC, the rising trends of SU and SU-SDA are notably more pronounced in the Arabian region (highlighted within red circles), consistent with the spatial distribution of the monthly minimum fAOD value trends. Fig. 8(e)–(h) shows the frequency changes in the occurrence of higher fAOD values (>1 standard deviation above the average) between the years 2000–2003 and 1995–1999. The higher ratio values in Fig. 8(e)–(h) (exceeding 50%, marked by black boxes) demonstrate the evident increase in higher fAOD values across western USA Conversely, across all four datasets, the occurrences of higher fAOD values in the Amazon region significantly decrease (below 50%, shown by black circles). It is noteworthy that in the Amazon region, the frequency of lower fAOD values (<1 standard deviation below the average) noticeably increases in SU-SDA, SU, and

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ORAC [refer to the Supplementary Material, Fig. S5(e)–(h)]. Furthermore, the increase in frequency of higher fAOD values is magnified, primarily in the Arabian region, northern India, and western China.

IV. Discussion and Conclusion

Three new SDA-based global daily fAOD datasets from 1995 to 2003 with $1^\circ \times 1^\circ$ spatial resolution are presented in this study. Compared with the AERONET data, the application of SDA can improve the performance of ADV–SDA and SU–SDA fAOD, with a higher R value and more retrievals falling within the EE envelope. However, when applying SDA to the ORAC product, the ORAC–SDA fAOD accuracy is considerably worse than that of the official ORAC fAOD; R decreases from 0.62 to 0.58 and RMSE increases from 0.14 to 0.16. Through validation, we found that SU–SDA fAOD has the strongest correlation with AERONET fAOD over global-scale validation ($R = 0.72$ and 65.36% retrievals within the EE envelope). From a spatial perspective, SU–SDA captures hotspots well in developing countries, such as the Central African Republic, India, and China. The notably high fAOD in central South Africa suggests that the sources of fine aerosols were biomass from the burning of forested areas and oil–gas production fields [46]. It should be noted that the Niger Delta, which had high SU–SDA fAOD, has the greatest amount of gas flaring in the world.

The SU–SDA retrieval showed a large decrease in fAOD in central South America from 1995 to 2003. However, significant increasing trends were identified in India and China. Although previous studies have indicated that anthropogenic emissions in China increased in the 2000s [47], this study further illustrates that these emissions have grown rapidly since the 1990s. Seasonally, the SU–SDA results indicate that fAOD presented more statistically significant positive trends in winter than in other seasons in Australia and Mexico. In North China, the fAOD in summer was higher than that in other seasons. Although secondary fine particles (such as $PM_{2.5}$) are easily produced by photochemical reactions under high temperatures and humidity during the summer months [48], [49], their sources of aerosol humidification and coagulation can be other major contributors [50]. We found that higher fAOD values were accelerating in western USA, the Arab region, northern India, and western China from 1995 to 2003 while decreasing in the Amazon region.

As an entirely different method to calculate fAOD, compared with the general satellite-based method, SDA requires only spectral AOD values at multiple channels. However, high-accuracy and stable spectral AOD at each channel cannot be easily obtained [51]. The uncertainty in the AOD can cause large errors in the AE (α) and AE derivative (α') calculations. This means that AOD at 0.55 μm may be accurate, but it is not reliable at 0.67 and 0.87 μm. This could cause high uncertainty in FMF estimations. Thus, further efforts should focus on the accuracy of AOD retrievals for each channel. Current satellite-based AOD retrievals have several important limitations, such as surface reflectance determination, aerosol mode setting, and cloud contamination; integrating them with artificial intelligence [52] may be an effective solution to overcome the limitations of traditional methods.

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